

Leveraging open-access data in ChEMBL to explore emerging drug modalities



Emma Manners

Chemical Biology Services Team, EMBL-EBI

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Freely available resources for drug discovery

Chemical Biology Services from EMBL-EBI:



ChEMBL

*Bioactivities,
drug/candidate
information*



SureChEMBL

*Structures from
patents*



UniChem

*Linking
databases via
InChI*



ChEBI

*Chemical ontology,
and chemical
registration db for
biological dbs*



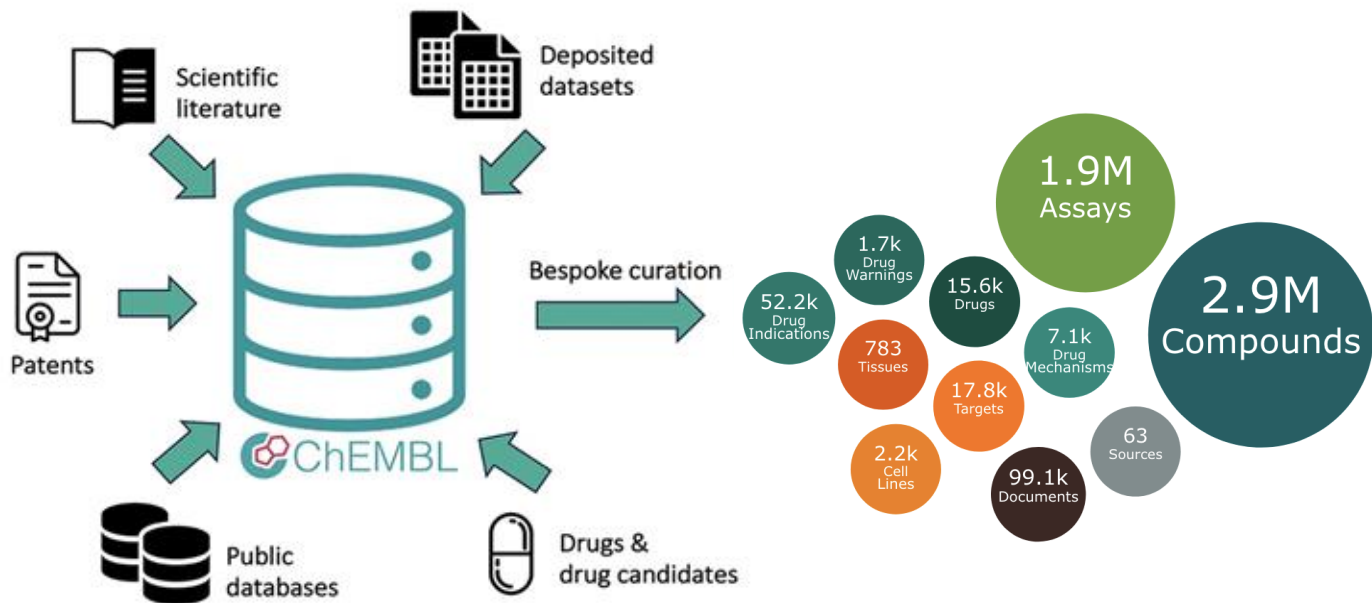
OPSIN

*Convert IUPAC
names to
structure*

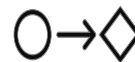
New!

We provide websites, database downloads, webservice, help desk/bug reports, training

ChEMBL offers precompetitive drug discovery data



Explore:



Compound-target interactions through bioassays



Extensively curated drug and clinical candidates



ADME, toxicity, physicochemical properties

Diverse data types; broad coverage of compounds & targets; FAIR

Diverse and emerging modalities are captured in ChEMBL

Oligonucleotides



Givosiran-GalNAc (Porphyrias, Hepatic)
Lumasiran (Primary hyperoxaluria type 10)
Patisiran (Amyloid Neuropathies, Familial)

Proteins/peptides



Semaglutide
Diabetes mellitus and obesity

Cell-based therapies



Brexucabtagene autoleucel (CAR-T)
Mantle Cell Lymphoma (2020)

Antibody drug conjugates



Gemtuzumab ozogamicin
Acute Myeloid Leukemia (2000)

Onasemnogene abeparvovec
First gene therapy for Spinal Muscular Atrophy in those under 2 yrs

Enzymes



Pegvaliase
Phenylketonuria

Vaccine component



SARS-CoV-2 virus spike protein
COVID-19

Biologicals

(mRNA) cancer vaccines



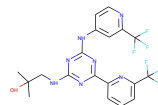
Biopepimut-S (Phase II)
Squamous Cell Carcinoma of Head and Neck

CRISPR-Cas therapies

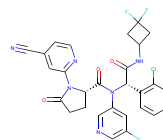


Exagamglogene autotemcel
Beta Thalassaemia and Sickle Cell Disease

Variant-selective drugs



Ivosidenib
IDH1 R132X
Enasidenib
IDH2 R140Q, R172X
Acute Myeloid Leukemia

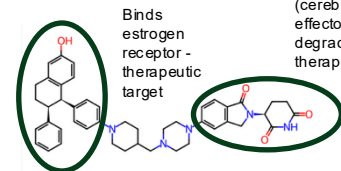


Precision medicine

Targeted protein degraders



Vepdegestrant (Phase III)
Breast Neoplasms



Binds E3 Ligase (cereblon), the effector causing degradation of the therapeutic target

Probody-drug conjugate



Praluzatamab ravtansine (Phase II)
Cancers
Masked antibody that targets "undruggable" CD166.

Emerging modalities

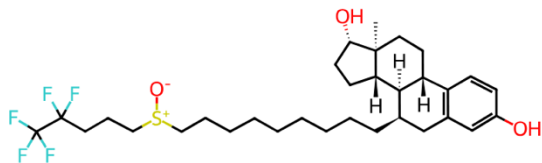
Novel drug delivery and activation features

(Targeted) protein degraders (TPDs) *remove* pathogenic proteins

Degraders

Monofunctional

- binding to a pathogenic protein causes downstream degradation



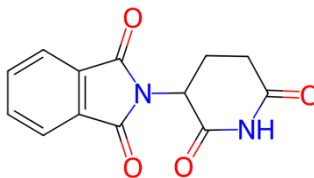
Fulvestrant

Selective estrogen receptor degrader (SERD)

Molecular glues

Conditionally bifunctional

- binding to an endogenous degrader alters its affinity - neosubstrates are bound and degraded

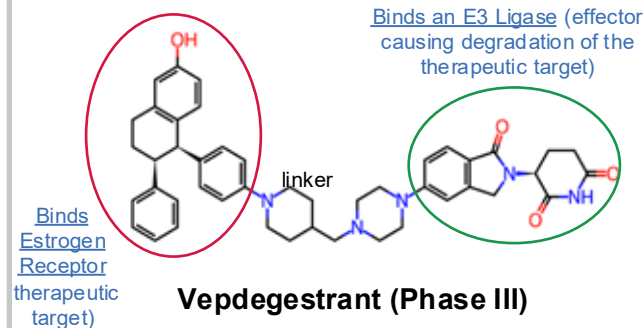


Thalidomide

Targeted protein degradation (TPD)

Heterobifunctional

- binds a pathogenic protein target and a degradation effector – places these in proximity leading to degradation of the *target*

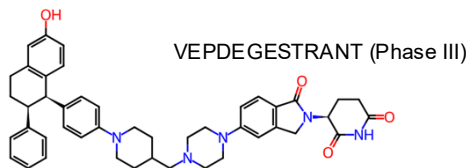


Vepdegestrant (Phase III)

Potential for “undruggable targets”
(catalytic mechanism, weaker affinity)

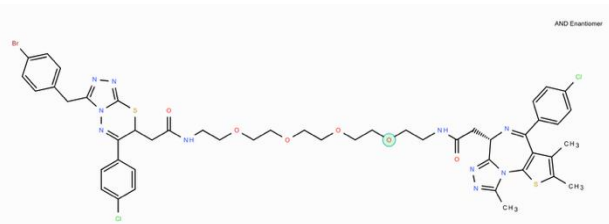
A suite of TPD platforms have been developed

Proteasome pathway



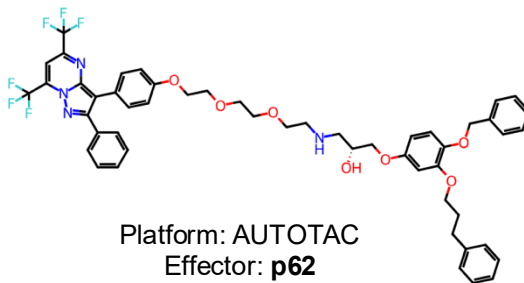
VEPDEGESTRANT (Phase III)

Platforms: PROTAC/SNIPER/AUTO-RAPTAC
TRAFTAC/AATAC
Effector: **E3 ligase**

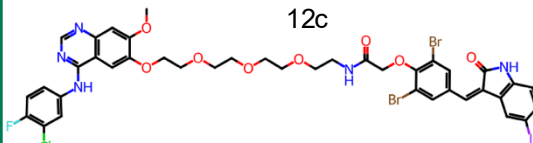


Platform: BYETAC
Effector: **Rpn-13** (direct proteasome degradation, bypasses E3 ligase)

Lysosome pathway

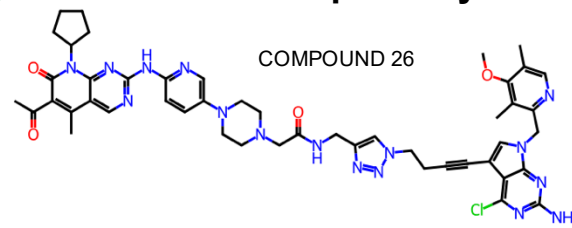


Platform: AUTOTAC
Effector: **p62**



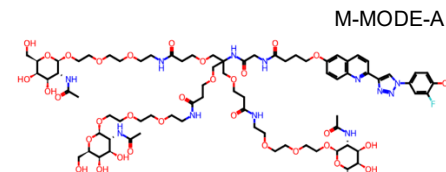
Platform: ATTEC
Effector: **LC3**

Heat shock pathway



COMPOUND 26

Platform: HEMTAC
Effector: **HSP90**



M-MODE-A

Platform: LYTAC
Effector: **internalising receptor**

Ontological coverage is necessary

Routine extraction of TPD-activity data

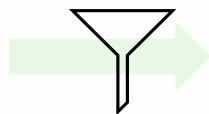


Zainab Ashimiyu-Abdusalam

1. Accessions

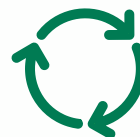
*Q96SW2
P40337...*

Curated
UniProt
accessions
for “effectors”



Filter:
Target type
Document terms

Accession-
based
activities

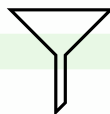


Refinement, ongoing
curation of legacy
data

2. Keywords

*PROTAC
HEMTAC...*

Curated TPD-
related assay
keywords



Filter:
Target type

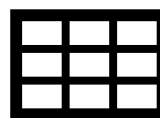
Degradation
terms

*Induction of *
degradation...*

Keyword-
based
activities

Light
clean-up

Combined
TPD-related
activities



TPD dataset

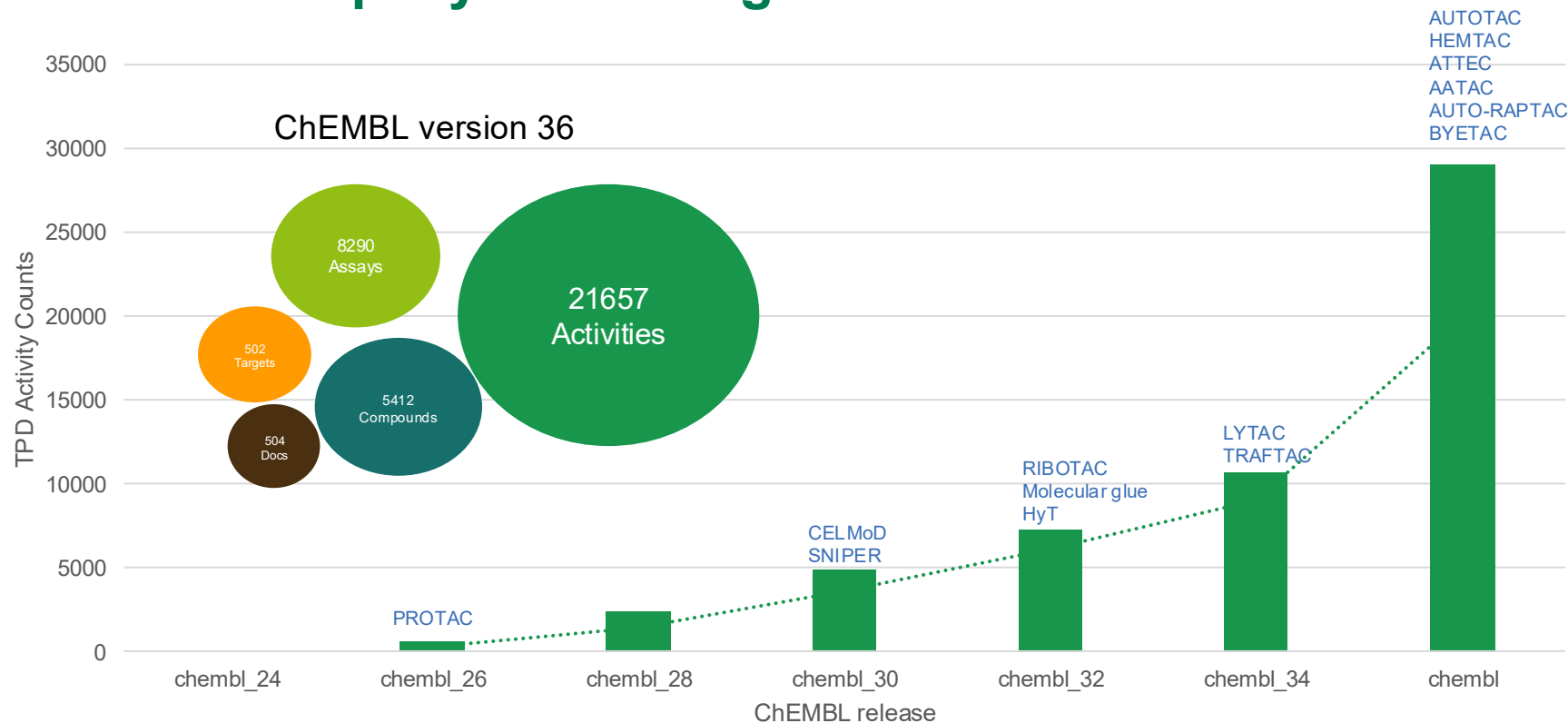
- assays
- activities
- targets
- structural information

**Coming
soon!**

A new
“**Modality**” flag
for ease of
access to TPD
data

Simple, low noise, adaptable

TPD data is rapidly increasing in ChEMBL



Targeted Protein Data in ChEMBL

Vepdegestrant (Phase III, TPD) can be profiled using ChEMBL

ADMET data

- **Bioavailability (mouse, 1-3 mg/kg, 24hrs):** 31-59 %
- ***in vivo* clearance (mouse, 1 mg/kg, 24 hrs):** 21.9-35 ml/min.kg (po or iv)
- ***in vitro* clearance:** 3 uL/min (human hepatocytes)

[CHEMBL6140274, CHEMBL5325057, CHEMBL5554434, CHEMBL5554443, CHEMBL6086130]

in vivo efficacy



Mouse xenograft models:
Antitumour activity against ERalpha WT and variant-driven growth.

[CHEMBL5325110, CHEMBL5325119]

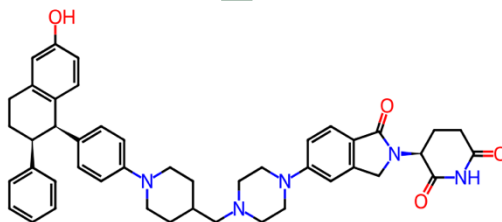
X-refs



CHEBI: 747473



SureChEMBL: 20211870



Vepdegestrant

ID: CHEMBL5095210, **Synonym:** ARV-471
Indication: Phase III (Breast cancer)

Compound properties

Calculated

- **MWt:** 723.92
- **Ro5 violations:** 2
- **Aromatic rings:** 4



Experimental

- **LogD:** 5.7 (rpHPLC), 4.6 (shake flask)
- **Solubility:** 275 ug/mL (pH 5, UPLC)

[CHEMBL6140262, CHEMBL6131311, CHEMBL6140284]

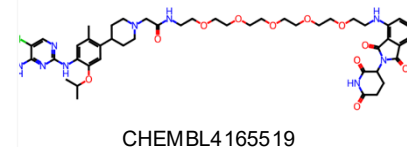
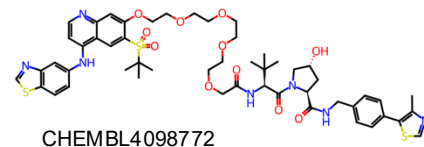
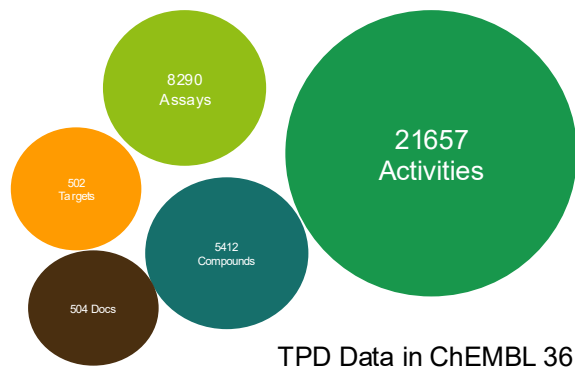
On-target cellular activity



DC50: 0.4-2 nM (ERalpha, MCF7 cells)

[CHEMBL5325054, CHEMBL5512778]

ChEMBL is a source of diverse preclinical TPDs



Top TPD targets: *

Degradation effector	Cereblon	VHL	LC3b	cIAP1
Pathogenic protein	AR BRD4 BTK	AR BRD4 EGFR	EGFR NLRP3 PDEdelta	ABL-BCR AR AhR

* by activity count

Compound properties

Av MWT 917 (vs. 443)

Ro5 violations 2 (vs <1)

Challenging targets

6% activities for variants (vs. 3%)

Undrugged targets e.g STAT3

Viral targets:

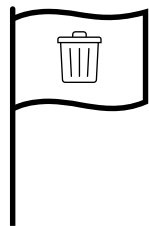
Influenza A virus, Severe acute respiratory syndrome coronavirus 2

Summary



ChEMBL

Open-access bioactivity data, including for emerging modalities.



TPD Flag

Rapid extraction of TPD data with our new “Modality” flag.



Explore degrader-target properties

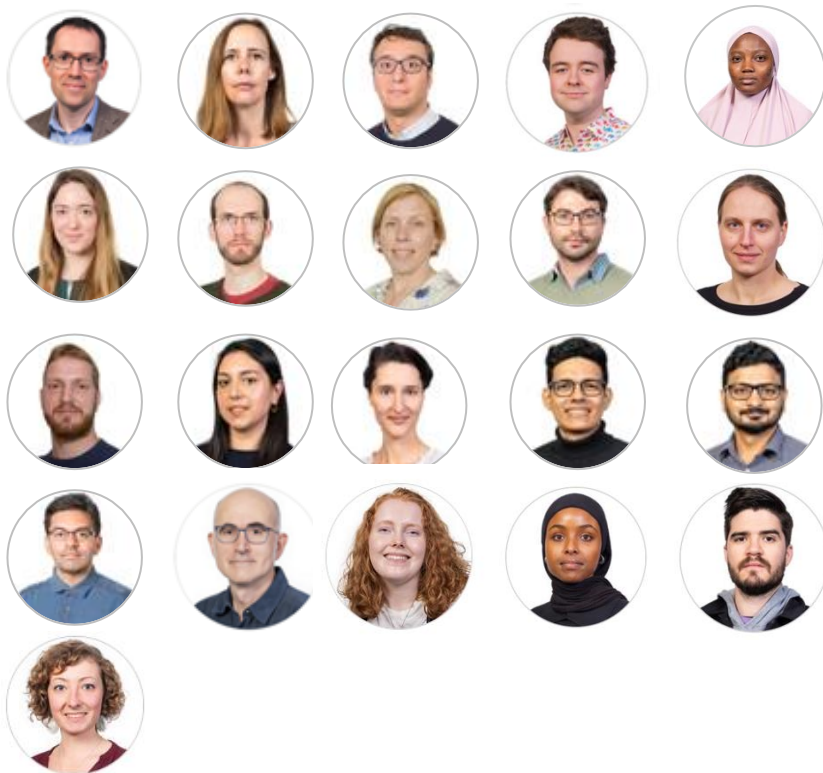
Degrader-target interactions, properties, pharmacokinetics, and safety.



Towards “AI-readiness”

Towards “AI-ready” datasets and “AI-assisted” curation

Chemical Biology Services



wellcometrust
Strategic Award



Get in touch!



*chembl-
help@ebi.ac.uk*



Public event/Ext Seminar event | Conference

2nd ChEMBL User Group Meeting



Scan for
more
details!